



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 2203: Software Engineering

Semester 01, 2024/2025

PROJECT PROPOSAL

ALL-IN-ONE FINANCIAL PLATFORM

Lembaga Kemajuan Pertanian Kemubu Kelantan (KADA)

Team Name: **Tech EcoPioneer**

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1. Introduction

Need (N)

Traditional cooperative management often faces inefficiencies, such as cumbersome manual processes for savings, loan applications, and document management. These inefficiencies hinder user experience and limit financial inclusion, especially in underserved communities. Research highlights that digital transformation in cooperatives addresses these issues by automating workflows, reducing the time and effort for members to access services like loans and savings, and ensuring safer and more secure financial transactions. The Koperasi KADA Online Platform System addresses the growing demand for a streamlined, efficient, and accessible cooperative management solution. By providing a digital platform, the system ensures convenience, transparency, and improved member experience, backed by features that automate critical tasks and reduce administrative inefficiencies.[1][2]

Approach (A)

The Koperasi KADA Online Platform System is a cloud-based web application designed to optimize cooperative management. It offers an intuitive account dashboard for members, enabling features like registration, automated savings through recurring payments, and seamless loan applications with a built-in calculator. Advanced tools like document auto-validation and OTP-based login through WhatsApp enhance security and user convenience by offering a seamless customer experience. Additional functionalities, such as transaction history, email notifications, chatbot support, and an accessibility-friendly design, ensure inclusivity and efficiency in managing cooperative operations. [3]

Benefits (B)

The Koperasi KADA online platform system provides a host of benefits that include enhanced user convenience by smoothening the processes for member registration, saving account applications, recurring payment options for automated savings, and secure OTP login via WhatsApp. Its loan calculator and transaction history features promote financial transparency, while automated document validation reduces administrative delays. The account dashboard and report summary provide centralized management for members to make informed decisions. Besides that, the chatbot/FAQ and customer support ticket systems guarantee timely assistance to improve user satisfaction. Accessibility support enables the inclusion of users with disabilities, while email notifications will keep the members updated, building trust and engagement. All these put together optimize cooperative management by offering a secure, efficient, and user-friendly experience.[4][5]

Competitor (C)

The Koperasi KADA online platform system is in competition with the advanced digital platforms of commercial banks for savings and loans, although the latter may not be very accessible to the poorer or unserved communities. For instance, Touch 'n Go eWallet and Boost provide users with easy saving and spending solutions; however, these do not possess unique, co-op-centric functionalities, like automatic recurring savings, customized loan services, or both. Digital microfinance platforms, such as Fundbox and Mynt, provide loan calculators for small-scale financial needs but do not have an extended functionality related to the community. Credit union management systems, like CUInsight or Fiserv, offer similar services of member management and automated workflows. However, KADA comes with unique features such as WhatsApp OTP login, accessibility support, and seamless ticketing systems, catering specifically to the operations of a cooperative and ensuring inclusivity and technology advancement.[6][7]

2. Existing Systems

The initial system of KADA Koperasi system is a manual system. The processes such as memberships registration, money transactions, and other operations are managed by filling the form manually and all the data stored in a paper. Thus, there are some issues exist in the system.

Issues with KADA Koperasi System

A) Lack of Integration

- The system may not integrate well with other tools or platforms, such as accounting software or banking systems
- Increase workload and error, causing inefficient workload
- **Evidence:** All process such as data transfer is undergone manually

B) Security Concerns

- Due to manual process of KADA Koperasi system, the security features are limited, causing the system is vulnerable to unauthorised access and data breaches.
- May cause leaked of users' personal information and financial fraud
- The impact is loss of trust among members and financial loss
- **Evidence:** The case related to financial fraud or data theft reported by cooperatives using similar systems.

C) Lack of Accessibility

- The system may not support mobile access or apps or any online features, making it inconvenient for members or administrators to access data.
- Reduced efficiency and inclusivity for members in remote locations
- **Evidence:** Many cooperatives in rural areas rely on mobile-friendly platforms due to limited access to computers.

D) Static Member Engagement

- Members can only rely on physical visits or manual processes for inquiries, payments, or updates.
- Members cannot easily track their savings, loans, or cooperative decisions.
- **Evidence:** Transactions and updates require physical visits to the cooperative office, leading to delays and inefficiencies.

The similar system that exists similar to KADA Koperasi system, which is CUMIS (Credit Union Management Information System), a cooperative system that for financial operations like savings, loans, and member benefits, but more focus on regulatory compliance and financial transparency. However, there are some issues in the system.

Issues with CUMIS

A) Focus on Financial Cooperatives Only

- CUMIS is primarily focus on financial cooperatives only, thus lacking integration with non-financial operations like inventory or retail systems.
- Limits use in multi-purpose cooperatives.
- **Evidence:** Reviews from cooperative users indicate the system is restrictive for cooperatives involved in non-financial services or diverse operations.

B) Limited Accessibility

- Interface is complex and not beginner-friendly, making it difficult for non-technical users to navigate the system.
- More frequent error due to user confusion.
- **Evidence:** User complaints about unintuitive workflows are common in reviews, with staff requiring significant IT support to perform basic tasks.

C) Limited Engagement Features

- Lacks interactive or engagement tools beyond financial operations
- No features that keep them informed about cooperative updates.

- **Evidence:** No mobile apps or portals are available for real-time updates or feedback collection

Current Solution

A) Implement APIs and middleware to enable integration with other systems

- Integrate with platforms like QuickBooks or Xero for accounting.
- This is to streamline the workflows and reduced manual data transfer.

B) Implement advanced security measures

- Because the system is implemented online, advanced security measures can be added into the system.
- Encryption for data storage and transmission, two-factor authentication (2FA) for user access
- Protection against unauthorised access and data breaches

C) Create online website that can undergo process above and user-friendly

- The website can be access by computer users and mobile users to undergo process such as member registration, transaction and payment with user-friendly features such as multi-language settings and guide of the process.
- Allow accessibility for remote users and real-time access to the system.

D) Dynamic Member Engagement

- Features such as notifications, self-service portals, online payments, and event updates are added into the website.
- Digital tools make it easier for members to engage with the cooperative anytime and from anywhere.

Features	Koperasi KADA Manual System	CUMIS	Solution
Integration	Limited or none; requires manual data transfer between systems.	Focuses on financial cooperatives	Full API and middleware integration with accounting, banking, and inventory systems for seamless workflows.
Security	Basic and weak	Provides basic security features like password protection and encryption	Advanced encryption, two-factor authentication (2FA)

Accessibility	Manual, not available online	Interface is complex and not beginner-friendly	Can be access by computer users and mobile users and users-friendly
Member Engagement	Focused on financial transactions only.	Static features with minimal interaction.	Dynamic features like notifications, online payments, and self-service portals.

NAMA: MOHD ROSLAN B ISMAIL
NO. K/P: 740807-03-5479 **NO. PF:** 2273
NO. AHLI: 1579

Tuan/Puan,

PENGESAHAN PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023.

Untuk penentuan Juruaudit, kami dengan ini menyatakan bagi akaun tuan/puan adalah sebagaimana berikut:

MAKLUMAT SAHAM AHLI:

Modah Syer:	Modal Yuran:	Simpanan Tetap:	Tabung Anggota:	Simpanan Anggota:
RM	RM	RM	RM	RM

MAKLUMAT PINJAMAN AHLI:

Al-Bai:	Al-Innah:	B/Pulih Kenderaan:	Road Tax& Insuran:	Khas:	Al-Qadrul Hassan:
RM0.00	RM0.00	RM0.00	RM0.00	RM0.00	RM0.00

PENGESAHAN BAGI PENYATA KEWANGAN AHLI KOPERASI KAKITANGAN KADA KELANTAN BERHAD BAGI TAHUN BERAKHIR 30 JUN 2023

Saya **MOHD ROSLAN B ISMAIL** No. Ahli: **1579** mengesahkan bahawa Penyata Kewangan Koperasi Kakitangan KADA Kelantan Berhad bagi tahun berakhir 30 Jun 2023 adalah benar:

Tandatangan: _____
Tarikh: _____

☐ Setuju
☐ Tidak Setuju.....
(Nyatakan sebabnya)

Nota:
1. Sila tandakan (/) di salah satu kotak diatas.
2. Sekiranya pihak Koperasi tidak menerima sebarang maklumbalas daripada tuan/puan sehingga 30 Jun 2023, maka pengesahan penyata ini dianggap betul dan tuan/puan bersetuju.

Figure 1: Paper for Verification of Member Financial Statements

3. Proposed System

The proposed system is an all-in-one platform to streamline operations for users from different background. The system supports various functionalities. It ensures the efficient handling of savings, loans, and customer interactions while providing integration with third-party software and tools.

Features	Explanation
Member Registration	Employees can register as members by providing their details.
Saving Account Application	Members can apply for savings accounts to deposit money and set savings goals.

Recurring Payment Option	Members can schedule automatic payments for regular savings.
Loan Application	Members can apply for loans, submit documents, and track application status.
Loan Calculator	A calculator to calculate loan repayment amounts and schedules.
Report Summary	Provides summaries of financial activities for staff and directors.
Account Dashboard	Personalised dashboards showing account details, balances, and activities.
Email Notification	Automatic emails to inform users about updates, approvals, and reminders.
Document Management	Automatically checks and validates documents uploaded by users.
Chatbot/FAQ	A chatbot to answer common questions and assist users.
Customer Support Ticket	Members can raise tickets for issues, and staff can resolve them.
Accessibility Support	Designed to be usable by people with disabilities, following the government accessibility standards.
OTP Login	Users can log in securely using one-time passwords sent to their WhatsApp.
Transaction History	Shows all deposits, withdrawals, and loan repayments in one place.

Table 1: Proposed System Features

Roles	Capabilities
New User	• Register as member by providing personal details • Upload required documents • Use chatbot to inquire frequently asked questions
Member	• Open savings account and make deposit • Apply for loans and manage repayments • Set up recurring payments for savings • Check transaction history and dashboard

Staff	• Approve or reject member registrations and loan applications. • Validate documents. • Answer support tickets. • Create reports for management.
Board of Directors	• Review detailed financial reports. • Approve large loans. • Monitor overall system performance and suggest improvements.

Table 2: User Roles and Capabilities

Aspect	Details
Third-party Integration	• Connects with payment gateways for savings and loan repayments • Uses VerifyHubs API for OTP • Integrated with email systems for notifications
Scalability	The system can handle more members and transactions as the organization grows.
Security	Ensures secure logins, encrypted data, and compliance with data protection standards.
Ease of Use	User-friendly design for members, staff, and directors with accessibility features.

Table 3: System Integration and Details

4. Software Process Model

For this project, the Agile Software Development Model was chosen because it provides the flexibility and iterative approach required to address KADA's evolving needs. Agile fosters continuous collaboration between the development team and stakeholders, allowing KADA staff and management to provide regular feedback throughout each sprint. This iterative approach helps to improve requirements, ensuring that important functionalities like member registration and loan applications are tightly aligned with user needs. Agile enables the delivery of functional modules in short, manageable cycles, resulting in early deliverables that users can test and confirm. This not only improves user satisfaction, but it also decreases the possibility of adopting ineffective features. Furthermore, Agile's emphasis on short sprints and frequent reviews enables early detection and resolution of issues, reducing the likelihood of delays or costly errors. Hence, Agile is ideal for developing a robust and user-centric system

for KADA because it encourages adaptation, incremental delivery, and stakeholder interaction.

Agile development divides the Software Development Life Cycle (SDLC) into multiple sprints, with iterative activities in each phase. Below are the detailed activities mapped to Agile SDLC phases:

1. Planning

- Define project scope (goals, deliverables, timelines, and budget)
- Conduct stakeholder interviews and gather initial input
- Feasibility analysis (technical, financial, and operational feasibility)
- Outline roles, responsibilities, timelines, and objectives
- Define project risk management plan
- Conduct market research to analyse competitors, trends, and user needs
- Set up initial communication plan and reporting schedules

2. Requirements

- Create functional and non-functional requirements
- Conduct requirements workshop with stakeholders to refine requirements
- Prioritise requirements based on business value and technical constraints
- Create user stories and acceptance criteria
- Conduct surveys (user feedback, market trends)
- Review and refine requirements after feedback from stakeholders
- Obtain stakeholder sign-off on requirements

3. Design

- Design system architecture, database design
- Create wireframes and UI mockups
- Low-level design (detailed component design, API specifications)
- Design system security measures (authentication, data encryption)
- Conduct review session with stakeholders to confirm and adjust designs
- Create a prototype for user testing

4. Development

- Set up development environment (tools, servers, repositories)
- Develop backend features (database, server-side logic, APIs)
- Develop frontend features (UI components, client-side logic)
- Implement third-party integrations (payment gateways, APIs)
- Implement system security measures (encryption, authentication)

- Integrate backend and frontend
- Perform unit testing during development
- Conduct code reviews
- Complete integration of all features and fix bugs

5. Testing

- Create and review detailed test plans
- Write test cases and test scripts
- Perform unit testing and debugging
- Perform system and integration testing
- Conduct User Acceptance Testing (UAT) with stakeholders
- Log and resolve bugs and issues found during testing

6. Development

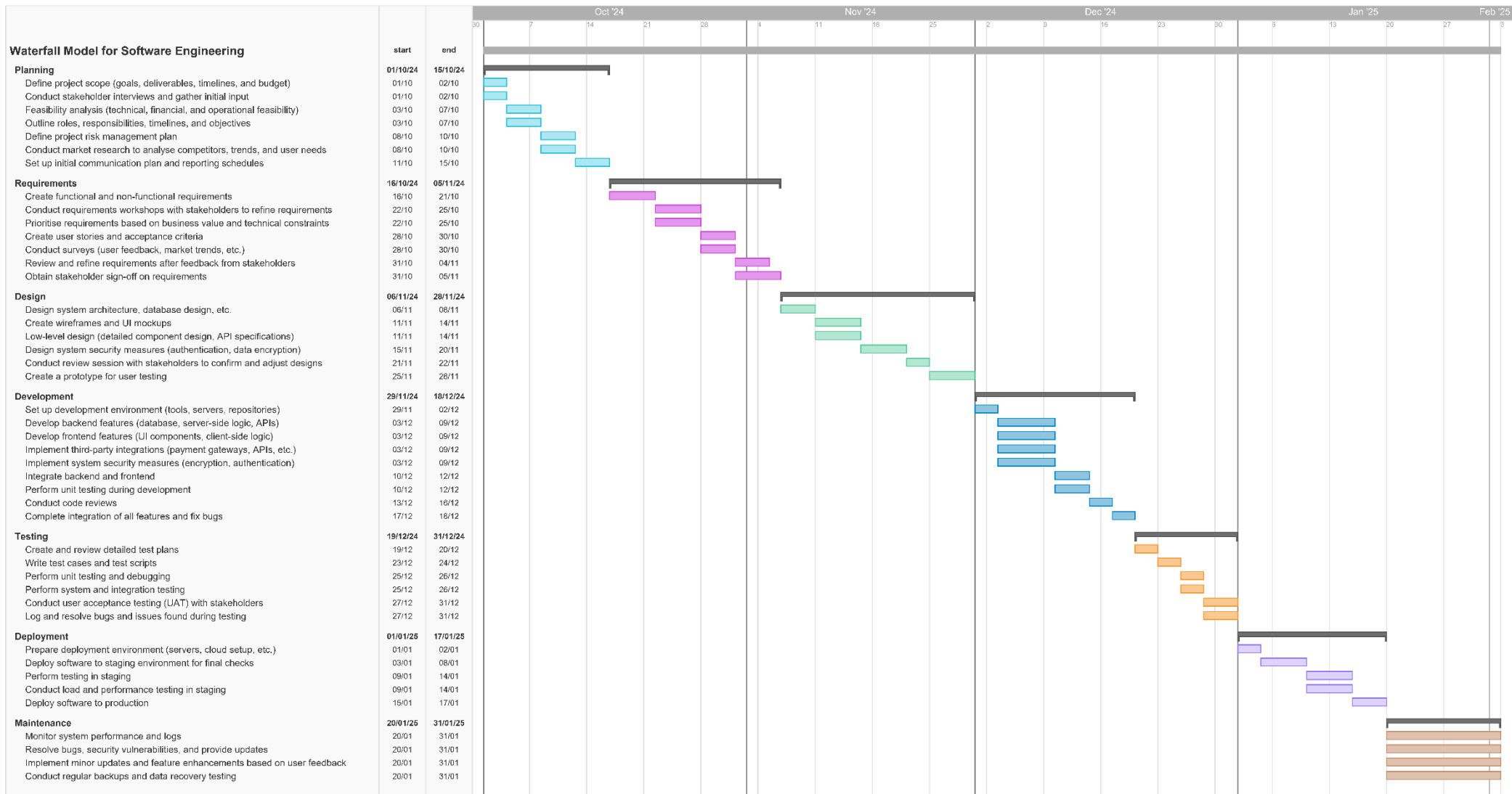
- Prepare deployment environment (servers, cloud setup)
- Deploy software to staging environment for final checks
- Perform testing in staging
- Conduct load and performance testing in staging
- Deploy software to production

7. Maintenance

- Monitor system performance and logs
- Resolve bugs, security vulnerabilities, and provide updates
- Implement minor updates and feature enhancements based on user feedback
- Conduct regular backups and data recovery testing

Phase	Duration	Deliverables	Person in Charge
Requirement Gathering and Analysis	2 weeks	Software Requirement Specification	All members
		Feasibility report	All members
Design	3 weeks	System Architecture Diagram	Evelyn Goh Ng Yu Hin
		Entity Relationship Diagram	Lau Yan Kai Dheshieghan
		Wireframes and prototypes for the user interface	All members
Development	5 weeks	Functional modules	Chew Chiu Xian Evelyn Goh Lau Yan Kai
		Source code	All members
		Updated sprint backlog	Dheshieghan Ng Yu Hin
Testing	3 weeks	Test Plan and Test Cases	Evelyn Goh Lau Yan Kai
		Test Summary Report	Dheshieghan Ng Yu Hin Chew Chiu Xian
Deployment	2 weeks	User manuals and training materials	Dheshieghan Evelyn Goh Lau Yan Kai
		User access credentials	Ng Yu Hin Chew Chiu Xian
		Deployment report	All members
Maintenance	Ongoing	Maintenance logs	Dheshieghan
		Feedback reports	Evelyn Goh
		Backup and recovery logs	Chew Chiu Xian
		Updated system documentation	Lau Yan Kai Ng Yu Hin

5. Project Schedule



6. References

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Appendices

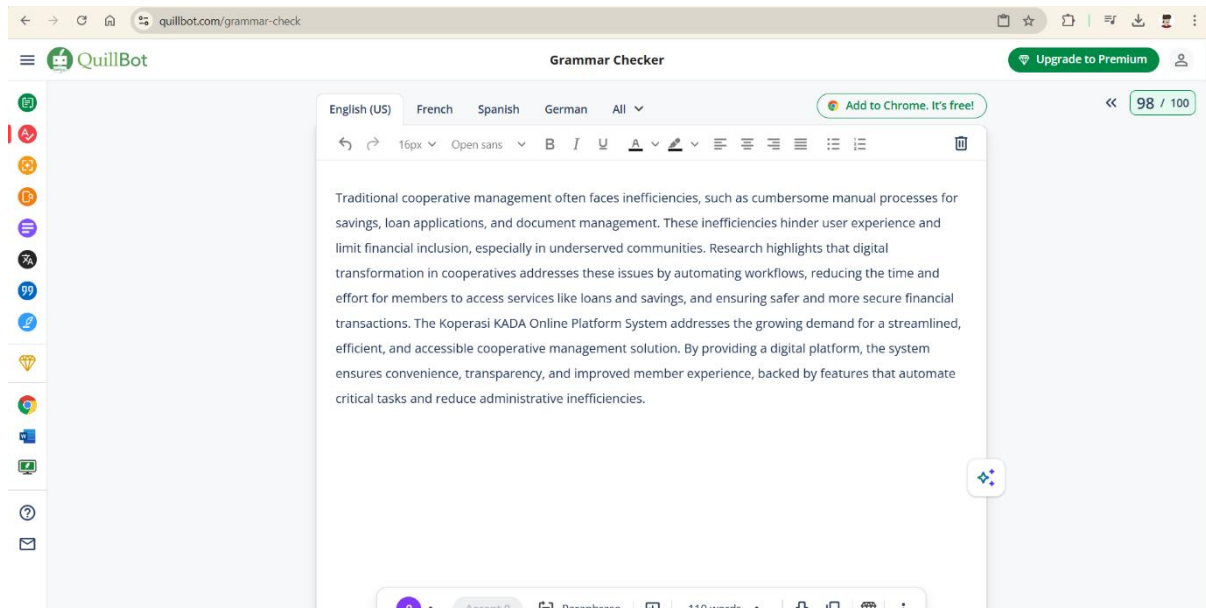


Figure 2: Screenshot Quillbot Grammar and Spelling Error Check

URL: <https://quillbot.com/grammar-check>

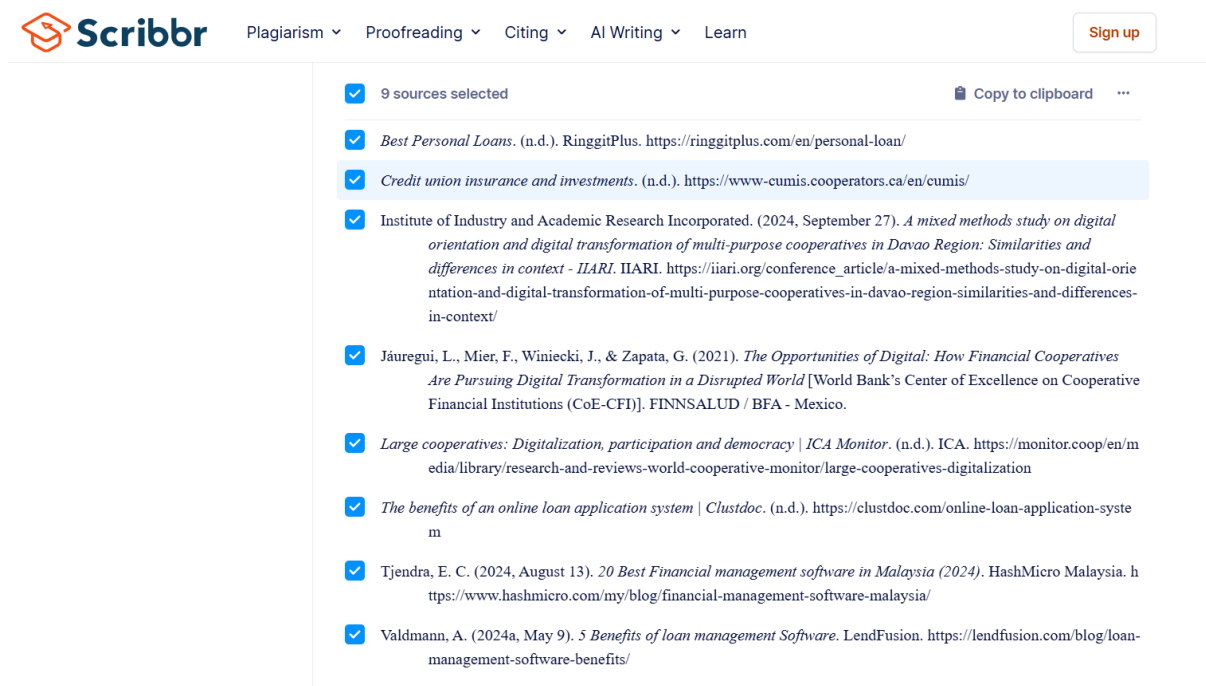


Figure 3: Screenshot Scribbr for Citation

URL: <https://www.scribbr.com/citation/generator>

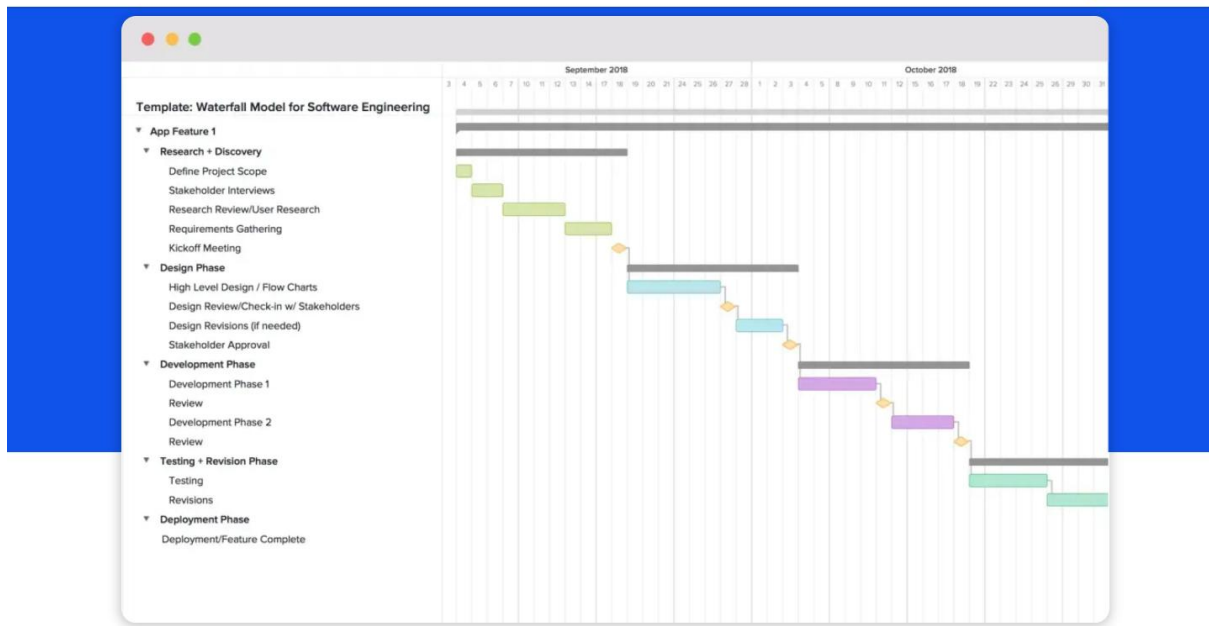


Figure 4: Screenshot teamgantt for Project Schedule

URL: <https://www.teamgantt.com/>

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